GitHub Actions Workflow – Complete Meaning and Step-by-Step Explanation

This document explains the complete meaning of a GitHub Actions workflow YAML file. Each section explains what GitHub does internally when a workflow runs using a GitHub-hosted runner.

Workflow YAML File

```yaml
name: GitHub Hosted Runner Demo

on:
  push:
    branches:
      - main
  workflow_dispatch:

jobs:
  run-python-job:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout repository code
        uses: actions/checkout@v4

      - name: Show runner operating system
        run: uname -a

      - name: Check Python version
        run: python3 --version

      - name: Run Python file
        run: python3 app.py
```

1. Workflow Name

The line **name: GitHub Hosted Runner Demo** defines the workflow name displayed inside the GitHub Actions dashboard.

2. Trigger Section

The **on:** section defines when the workflow starts. The workflow runs automatically whenever code is pushed to the **main** branch.

3. workflow_dispatch

This enables the manual **Run Workflow** button inside GitHub Actions.

4. Jobs Section

The **jobs:** section contains all tasks executed by GitHub Actions. A workflow may contain multiple jobs such as build, test, and deployment.

5. Job Identifier

The identifier **run-python-job** is the custom job name displayed inside workflow logs.

6. Runner Definition

The line **runs-on: ubuntu-latest** tells GitHub to create a fresh Ubuntu virtual machine called a GitHub-hosted runner.

7. Steps Section

The **steps:** section defines commands executed one-by-one inside the runner machine.

8. Checkout Repository

The action **actions/checkout@v4** downloads repository files into the runner environment.

9. Operating System Command

The command **uname -a** prints operating system details such as Linux kernel version and architecture.

10. Python Version Command

The command **python3 --version** checks whether Python is installed and shows the installed version.

11. Execute Python Program

The command **python3 app.py** executes the Python file stored in the repository.

12. Internal Workflow Execution

GitHub automatically creates the runner machine, downloads repository code, executes all commands, shows workflow logs, and finally destroys the virtual machine after completion.

Workflow Execution Architecture

Developer Pushes Code
GitHub Detects Push
Creates Ubuntu Runner

Downloads Repository
Runs Commands
Shows Logs
Deletes Runner

Conclusion

This practical demonstrates how GitHub Actions automates software execution using GitHub-hosted runners. The workflow file controls the entire CI/CD process through YAML-based automation instructions.